

Mithai Warriors

SemiRestore: AI restoration of degraded images
for semiconductor inspection

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2×

grayscale
restoration

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Degradation can hide defects before inspection

Observed input

128×128 or 256×256 grayscale
speckle + Gaussian noise; raw range may exceed
[0,1]

Required output

2× restored float32 image
sharp structure, low noise, no artificial patterns

Inspection depends on fine edges and texture; smoothing
can conceal the same evidence that restoration is meant
to recover.

What the solution must achieve

- Remove speckle without clipping valid extremes.
- Reduce Gaussian noise without blurring boundaries.
- Recover 2× spatial resolution in one model.
- Generalize to unseen semiconductor sources.
- Run fast enough for practical inspection workflows.

noise removal + 2× super-resolution + OOD robustness

One model adapts to every degradation

1 model

joint restoration

A single NAF-SR predicts a clean 2× residual instead of chaining separate denoising and upscaling models.

4 stats

image-conditioned

Mean, standard deviation, minimum and maximum modulate the network through zero-initialized FiLM.

2× skip

conservative residual path

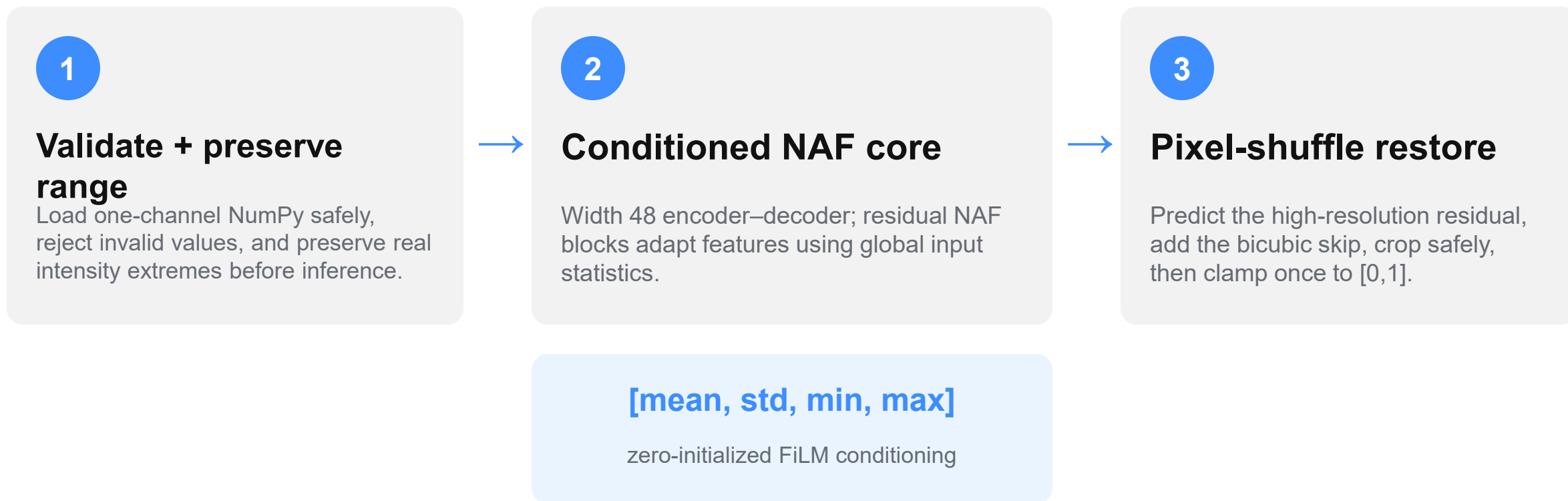
Bicubic carries stable low-frequency structure; the learned branch focuses on missing detail and noise correction.

0

public-test labels used

Model selection uses only organizer training pairs, locked validation-ID and texture-held-out pseudo-OOD evidence.

Safe input → conditioned NAF-SR → auditable output



9,111,684 parameters

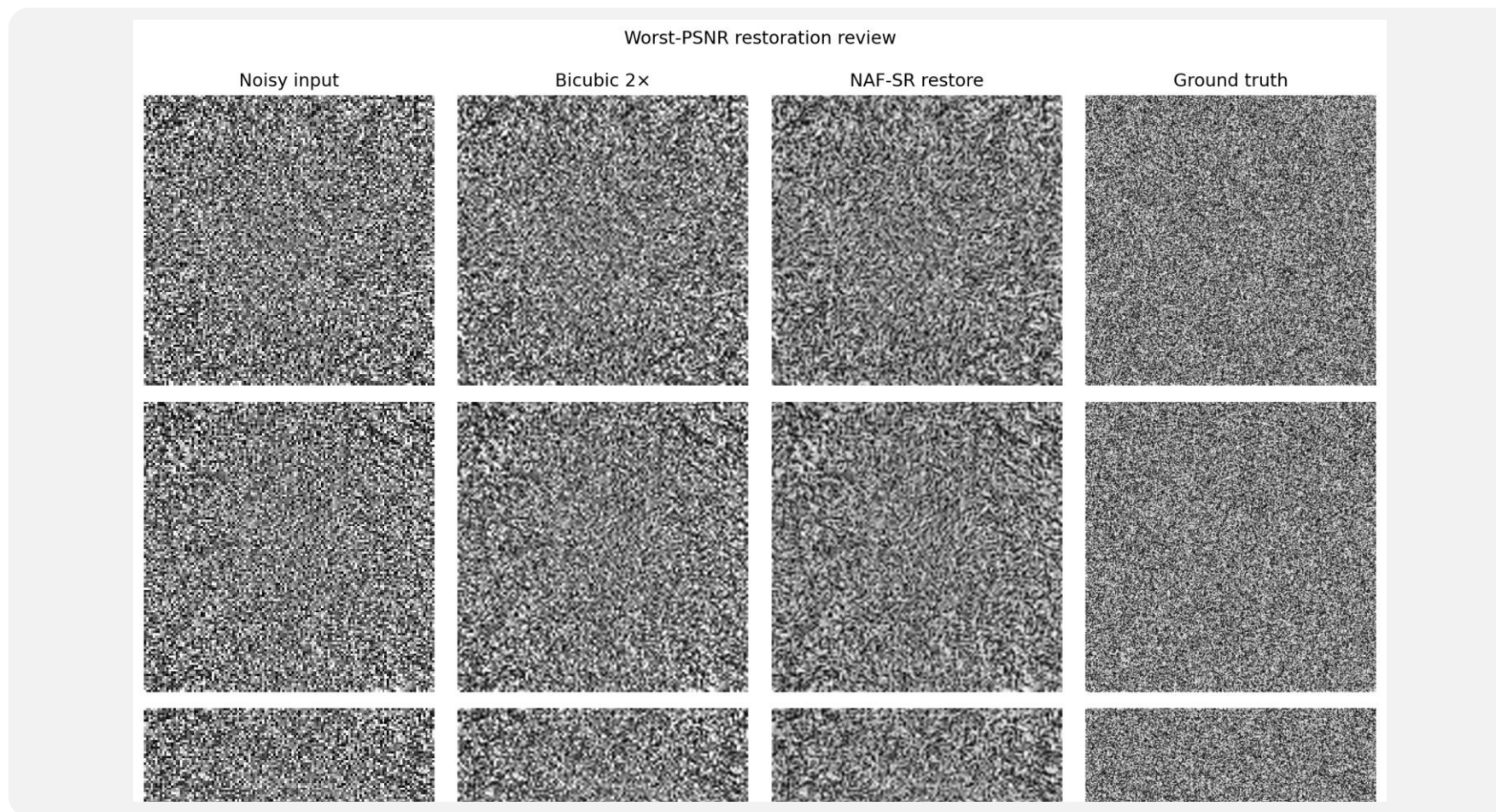
Charbonnier loss • real pairs • D4 geometry

Only OOD-gated improvements entered the final model

Design choice	Isolated change	Locked evidence	Decision
Texture-held-out split	12 clusters; entire outlying clusters held out	+5.02 dB vs bicubic on pseudo-OOD	CORE
+15% fitted synthetic	train-fitted noise and speckle profile	0 / 3 aggregate OOD metrics improved	REJECT
+statistics conditioning	FiLM from mean / std / min / max	PSNR +0.0071; SSIM +0.00152	KEEP

The architecture is compact; the novelty is evidence discipline—input statistics control restoration strength, and every change must improve locked OOD evidence without material ID loss.

≈ 5 dB gains survive worst-case visual review



Locked metrics

- Validation-ID 29.226 dB • 0.7766 • 0.2714
- Pseudo-OOD 25.251 dB • 0.6661 • 0.3530
- Bicubic gain +5.43 dB ID
- Bicubic gain +5.02 dB OOD

Cold external check

Noise profiles: +8.16 /
+4.10 dB
Clean-only: -2.18 dB

A100-tested, compact, and clean-clone verified

9.11M

parameters • 34.87 MiB checkpoint

1.890 ms

A100 compiled FP32 • batch 1

1,958 img/s

A100 compiled FP16 • batch 16

Eager FP32

Eager FP32 avoids a 20–23 s compile: 14.871 ms batch 1 and 542.61 images/s batch 16 on A100.

Package gates

400/400 outputs • 52 tests • complete A100 pip freeze

Fresh public clone passed checkpoint hash, output count/path/shape/dtype/range checks, default evaluator smoke, and notebook validation.

```
git clone → pip install → python evaluation.py INPUT_DIR OUTPUT_DIR → verify_submission.py
```

Clone. Run. Verify.

- STEP 1** Clone the public repository
- STEP 2** Install requirements-runtime.txt
- STEP 3** Run evaluation.py INPUT_DIR OUTPUT_DIR
- STEP 4** Verify all 400 restored arrays
- STEP 5** Optional Colab training walkthrough

Public repository

**github.com/
FaisalTabrez/Semicon**

Final model
conditioned_naf_sr

Checkpoint
273abd9d...158c28

Public code
and weights included

Ready for judge-side execution

Every claim is traceable to code, data, or literature

Restoration networks

NAFNet — Chen et al., ECCV 2022
EDSR — Lim et al., CVPRW 2017

Quality metrics

LPIPS — Zhang et al., CVPR 2018
SSIM — Wang et al., IEEE TIP 2004

Architectures and metrics were selected for restoration fidelity—not defect classification.

Data & implementation

- i4C / KLA AI restoration challenge statement (2026)
- KLA paired organizer dataset: 3,200 training samples
- Carinthia SEM Defect Dataset, Zenodo 10715190
- PyTorch 2.11, scikit-image and LPIPS
- Public code, model card and evidence reports in GitHub

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